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MILITARY STANDARD

SOLDERING OF ELECTRICAL CONNECTIONS AND PRINTED WIRING ASSEMBLIES, PROCEDURES FOR

TO ALL HOLDERS OF MIL-STD-1460(MU).

1. The following pages of MIL-STD-1460(MU) have been revised and supersede the pages listed.

New Page	Date	Superseded page	Date
iii	1 November 1973	iii	1 September 1973
v	1 November 1973	v	1 September 1973
ix	1 November 1973	ix	1 September 1973
1	1 November 1973	1	1 September 1973
3	1 November 1973	3	1 September 1973
6	1 November 1973	6	1 September 1973
7	1 November 1973	7	1 September 1973
9	1 November 1973	9	1 September 1973
15	1 November 1973	15	1 September 1973
27	1 November 1973	27	1 September 1973

2. The following page is to be deleted:

Page No.	Date
28	1 September 1973

3. Retain this notice page and insert before the table of contents.

4. Holders of MIL-STD-1460(MU) will verify that page changes and deletions indicated above have been entered. This notice page will be retained as a check sheet. This issuance, together with approval pages, is a separate publication. Each notice is to be retained by stocking points until the Military Standard is completely revised or cancelled.

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DEPARTMENT OF DEFENSE

WASHINGTON, D. C. 20301

Soldering of Electrical Connections
and Printed Wiring Assemblies, Procedure for

MIL-STD-1460(MU)

1. This military standard has been approved by the Picatinny Arsenal, Department of the Army and shall be used by that activity. All other military activities are required to employ this standard, where suitable.

2. Recommended corrections, additions, or deletions should be addressed to the Commanding Officer, Picatinny Arsenal, ATTN: SARPA-QA-N-Q, Dover, New Jersey 07801.

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The soldering procedures covered by this military standard are to be utilized for the soldering of highly reliable electrical connections in the electrical and electronic equipment of military systems. This military standard outlines the basic soldering procedures which are to be employed. It also graphically illustrates the accept/reject criteria for a variety of soldered connections. It is recognized that soldering connections exist other than those dealt with herein. It is intended that these connections shall be handled under the general provisions of this standard.

The use of methods other than hand soldering is acceptable provided the same quality standards are applied to the finished product and the finished product complies with the requirements specified herein.

This military standard includes some information obtained from Specifications MIL-S-45743C(MI), MIL-S-46844A(MI), and NASA publication NPC 200.

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1. SCOPE

1.1 Scope. - This standard prescribes general requirements for the soldering of electrical connections and printed wiring assemblies. (See 6.1 and 6.3.)

2. REFERENCED DOCUMENTS

2.1 Government documents. - The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification.

SPECIFICATIONS

Federal

O-E-760	Ethyl Alcohol
QQ-S-571	Solder; Tin Alloy, Lead-Tin Alloy; and Lead Alloy
TT-I-735	Isopropyl Alcohol

Military

MIL-I-7444	Insulation Sleeving, Electrical, Flexible
MIL-F-14256	Flux, Soldering, Liquid (Resin Base)
MIL-I-22076	Insulation Tubing, Electrical, Nonrigid, Vinyl, Very Low Temperature Grade
MIL-I-22129	Insulation Tubing, Electrical, Polytetrafluoroethylene Resin, Nonrigid
MIL-I-23053	Insulation Sleeving, Electrical, Heat Shrinkable; General Specification for
MIL-S-45743C(MI)	Soldering, Manual Type, High Reliability Electrical, Electronic, Instrument, Communication, And Radar For Aerospace, And Control Systems, Procedures For
MIL-S-46844A(MI)	Solder, Bath Soldering of Printed Wiring Assemblies, Automatic Machine Type

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Component Lead	The wire, solid or stranded, which extends from and serves as a connection to a component.
Conduction Soldering	Method of soldering which employs a soldering iron for transfer of heat to the soldering area.
Disturbed Solder Connection	Unsatisfactory connection resulting from relative motion between the wire and terminal during solidification of the solder.
Epoxy Potting Compound	A nonconductive, organic plastic used for encapsulation of components and/or wires.
Excessive Solder	Unsatisfactory connections wherein the solder obscures the configuration of the connection.
Eyelet	A tubular metal part having both ends headed or rolled over.
Government Approved School for Hand Soldering	A school established at NASA installation to train agency personnel in Category III for NASA work at that installation, the soldering school conducted by Frankford Arsenal or equivalent.
Ground Support	Equipment used to prepare, test, and check launch vehicles and spacecraft prior to launch. Also used to launch, operate, and maintain vehicles and spacecraft in space. This includes static firing test equipment and other test equipment for ground tests whether or not operational during flight operations.
Hook Terminal	A terminal formed in a hook shape.
Perforated or Pierced Terminal	A terminal containing a hole through which leads or wires are placed before soldering.
Printed Circuit Pad	A terminal area on a printed circuit used for making electrical connections to the conductive pattern.
Resistance Soldering	Method of soldering by passing a current, which heats the soldering area, by contact with two electrodes (not for sensitive semiconductor components).

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4.2.2 Solder. - An Sn60 or Sn63 tin-lead solder, Form W, Type R, RMA, or S conforming to QQ-S-571 shall be used; Form B may be used for lead pretinning.

4.2.3 Flux. - Only rosin flux shall be used in all soldering operations. The use of liquid rosin type flux is permissible for applications such as the removal of excess solder from a joint by wicking (see 4.6.7.2) into stranded conductor wire, and for soldering with solid solder. Where dual usage of cored and solid solder is required, core flux and liquid flux must be compatible. Thinners compatible with the flux shall be used. Liquid rosin flux shall conform to MIL-F-14256, Type RMA, except that the copper mirror test (see 3.5 of MIL-F-14256) is not required and the resistivity of the water extract (see 3.2.6) shall be at least 45,000 ohm-centimeters.

4.2.4 Cleaning solvents. - Solvents to be used for the removal of grease, oil, and other dirt from the parts prior to soldering, as well as flux residues from the joint area, are divided into two groups as follows:

a) Non-flammable -

1) Tetrachloroethylene (perchloroethylene) - electronic grade.

2) Trichloroethylene - electronic grade.

3) Or any of the products approved by the procuring activity.

b) Flammable -

1) Ethyl alcohol conforming to O-E-760, Grade 1, class A or class B.

2) Isopropyl alcohol conforming to TT-I-735 - 99% pure.

4.2.4.1 Cleaning. - If parts and assembly can be totally submerged in the cleaning fluid without damage, this is the preferred method. In those cases where only brushing with the liquid cleaner is permissible, use only non-activated or mildly activated rosin core solder, where flux residues may be left in place. All residues from liquid fluxes should be removed.

4.2.5 Flexible insulation tubing or sleeving. - Except as otherwise required by the design, flexible electrical insulation tubing shall be either extruded vinyl plastic, polytetrafluoroethylene or heat shrinkable, as applicable. Polyvinyl chloride plastics shall not be used.

4.2.5.1 Extruded vinyl plastic tubing. - Extruded vinyl plastic tubing conforming to MIL-I-7444, Type I, may be used where exposure to temperatures in the range of -90° to +158°F will be encountered.

4.2.5.2 Tubing. - Polytetrafluoroethylene tubing which conforms to MIL-I-22129 may be used where exposure to high temperatures and high frequencies or both will be encountered. This tubing shall not be used in applications where the temperature exceeds 500°F. WARNING: When polytetrafluoroethylene resin compounds are heated above 400°F, small quantities of toxic vapors are emitted. Adequate exhaust ventilation shall be provided when polytetrafluoroethylene resin compounds are heated above this temperature.

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4.2.5.3 Heat shrinkable tubing. - Heat shrinkable tubing which conforms to MIL-I-23053 (Wep) may be used where shrink-fit tubing is required. Heat shrinkable tubing shall not be used where the application of heat will damage the soldered joint or component. Class 2, polytetrafluoroethylene heat shrinkable tubing shall not be used for this purpose.

4.2.6 Terminals. - Solder terminals shall readily accept solder according to MIL-STD 202, method 208 and require no presoldering preparation (e.g., tinning). Terminals may be plated and shall be provided with guide slots, holes, grooves, or similar features to ensure proper mating of the wire or wires with the terminals. Each terminal guide shall be located to facilitate application of solder. Terminals shall be of proper size to accommodate the conductors. Terminal slots and solder cups shall not be modified to accept oversize conductors, and conductors shall not be modified to accommodate undersize terminals. Terminals shall be stored in special sealed containers with a small bag of desiccant to prevent oxidation. The use of silver plated terminals on printed circuit boards shall not be permitted. Silicone base lubricants must not be applied on or near surfaces to be soldered.

4.2.7 Wire. - Wires shall conform to the applicable drawings and specifications and shall be capable of being stripped clean by means of the insulation strippers specified in 4.4.6. All wires, including lead wires, shall readily accept tinning and soldering, in accordance with MIL-STD-202, method 208.

4.2.8 Solderability. - The solderability of solid lead wires, terminals, and other termators which are normally joined by soft solder shall be determined in accordance with MIL-STD-202, method 208. They shall be tested in the "as received" condition and after aging according to paragraph 4.2, method 208.

4.3 Materials for soldering printed wiring. -

4.3.1 Materials. - Except when otherwise specified on detailed drawings, the materials utilized in soldering operations shall conform to the requirements of 4.3.2 through 4.3.6.

4.3.2 Solder. - The solder used shall be tin-lead conforming to composition Sn60 or Sn63 or Type S conforming to QQ-S-571.

4.3.3 Flux. - The soldering flux shall be noncorrosive and nonconductive rosin base flux (see 3) which has a melting point below the liquidus of solder. The flux shall have no harmful effect on the markings of components of the printed board and shall be readily removable with the solvent specified. The flux shall conform to MIL-F-14256 or be a product approved by the design activity.

4.3.4 Solvents. - An approved stabilized neutral, Ph 6.6 to 7.4, (as tested by a water extraction method) solvent shall be used which has the ability to degrease or remove rosin flux residue and other contaminates from printed circuit assemblies. Solvents shall conform to requirements of TT-I-735 Isopropyl Alcohol, Grade A, or be a product approved by the design activity.

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4.4.8 Soldering irons. - Soldering iron size (tip size and shape, voltage and wattage rating) and temperature shall be selected and controlled for optimum performance in relation to the work to be performed. (See 4.6.2.) Transformer-type solder guns shall not be used. Temperature controlled soldering irons are acceptable provided sensing of temperature is at the soldering iron tip.

4.4.9 Soldering iron holder. - A solder iron holder shall be provided for the soldering iron utilized. A cage type holder that leaves the soldering iron tip unsupported is preferred. The holder should be such that the iron handle is protected from rising heat. The holder should assist in maintaining proper iron temperature.

4.4.10 Soldering tools. - Tools shall not cut, nick, or in any way damage leads. Forked type tools generally referred to as soldering aids may be used, provided they are of the non-metallic type.

4.4.10.1 Thermal shunts. - Thermal shunts, or heat sinks, shall be used as necessary to protect heat-sensitive components such as semiconductors, crystal devices, meter movements, insulating materials, etc., from damage due to heat while soldering. Thermal shunts shall be of such material, size, shape, and design as to permit rapid application and removal with minimum interference to the soldering procedure and provide rapid heat removal from the area being soldered. Thermal shunts shall be held in place by suitable means, such as friction or spring tension, which will prevent damage to the surface and insulation of the wire and to the component being soldered. (See Figure 2.)

4.4.10.2 Anti-wicking tools. - Approved type anti-wicking tools marked with conductor gage size, shall be used when required (See Figure 2.)

4.4.11 Tools and materials for cleaning. - Tools and materials for cleaning component leads, contact areas, gold plating, and soldering iron tips shall be as specified herein.

4.4.11.1 Component lead cleaning. - Component leads shall be cleaned with an efficient cleaning tool. Such a tool can be constructed utilizing 1/2-inch tinned copper shielding braid mounted in a spring-type tool as pictured in Figure 3. Knives, emery cloth, sandpaper, and other abrasives shall not be used.

4.4.11.2 Soldered areas. - A medium-stiff natural or synthetic bristle brush or a lint-free industrial cleansing tissue, dipped into an approved solvent, shall be used to remove excess flux following solder solidification. Wire brushes, knives, emery cloth, sandpaper, and other abrasives that produce a harsh abrasive action shall not be used.

4.4.11.3 Gold plating. - For removing gold plating from solder areas, use a pencil-style white typewriter eraser. It shall be furnished with a brush attachment.

4.4.11.4 Care of soldering iron tips. -

4.4.11.4.1 Unplated tips. - Shall not be used.

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4.5.5.3 Termination of ungrounded shielding. - Braided shield that does not require a ground termination shall, at the point of termination, be insulated. The raw edge of the termination shall be insulated with insulation sleeving. The raw edge shall be folded or peeled back on the shield. The insulation sleeving shall be securely fitted over the termination, extending a minimum of 1/4 inch in both directions from the raw cut edge. (See Figure 19A.)

4.5.5.4 Termination of shielded single conductors in twisted pair or twisted shielded triple. - When terminating shielded single conductors of multi-conductor cable, the twisted shield shall be trimmed to approximately 1/2 inch from the breakout point and a piece of insulating tubing shall be applied over the shield and conductor insulation approximately 1/2 to 5/8 inch long covering the cut end of the shield and the shield breakout point. Sleeving MIL-I-23053 shall be shrunk to a tight fit. Breakout points may be staggered to prevent excessive diameter buildup of multiple conductor cables. (See Figure 19B.)

4.5.5.5 Termination of outer sheathed cables. - On outside type cables having a heavy outer sheath the overall shield shall be loosened by pushing back over the cable conductors. The shield shall then be folded back over the sheath approximately 1/2 to 5/8 inch. Cables will then have a piece of heat shrinkable insulating tubing applied over the folded back shielding and the sheath then shrunk in place. Tubing shall completely cover the shielding and extend approximately 1/4 inch beyond the cut end of the shield and 1/4 inch beyond the fold back point on the shield. (See Figure 20A.)

4.5.5.6 Termination of shields on cable harness. - Inside type cables or harness, having no outer sheath, will have a piece of heat shrinkable tubing applied over the wire bundle being approximately 3/4 inch back from the cut end of the shield and extending approximately 1/2 inch. The shield will then be loosened by pushing the shield back over the conductors. The loosened shield will then be folded back over the previously applied tubing and pulled taut. A second piece of insulating tubing will then be applied over the folded shield and extending from the fold point on the shield to at least 1/4 inch beyond both the previously applied tubing and the shielding. The tubing will be shrunk into place. (See Figure 20B.)

4.6 General soldering technique. (See 6.3.). -

4.6.1 Securing conductors. - The tinned wires and leads shall be formed and joined to the terminals as described in 4.5.4. Wires and leads shall be held snugly and rigidly to terminals in such a manner that there will be no motion relative to each other during the soldering operation and during cooling and solidification of the solder.

4.6.2 Heat application. - Apply the soldering iron tip to the connection area in such a manner that maximum heat will be transferred to the parts to be soldered and maximum protection will be afforded insulation and parts that will be adversely affected by excessive heat. Thermal shunts shall be used where required for the protection of insulation and heat sensitive parts. The connection area shall not be overheated. (See Figure 21A and the following table.)

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5.5 Rejects. - Defects shall be tagged and the type of deviation noted on the tag.

5.6 Rework. - Soldered joints, reworked to correct deficiencies, shall be one hundred percent reinspected for the deficiencies in question and for any other features that may have been affected by the reworking operation.

5.7 Inspection standards. - Figures 1 through 46 and any additional inspection provisions provided or approved by the procuring activity shall be used to aid in determining compliance with the requirements set fourth in this standard.

5.8 Qualification and certification. - The qualification and certification of soldering operators and soldering inspectors when required, (see 5.1.1.2), shall be in accordance with 4.8.

5.9 Work areas and environments. - Work areas and environments approval, when required (see 6.2), shall conform to 4.1.

6. NOTES

6.1 Intended use. - It is intended that the soldering procedures and techniques covered by this standard are to be utilized for the soldering of highly reliable electrical connections used for electrical and electronic equipment of military hardware and systems.

6.2 Ordering data. -

a) Title, number, and date of this military standard

b) If certification (see 4.8 and 5.8) of solderers and inspectors per this standard is not required, the appropriate contractual document shall indicate same.

c) If the work area requirements (see 4.1) of this standard are not required, the appropriate contractual document shall indicate same.

6.3 Soldering methods. - The use of soldering methods other than hand soldering is acceptable under this military standard provided the same quality standards are applied to the finished product and the finished product complies with the requirements specified herein. (See MIL-S-46844A(MI).)

6.4 Supersesssion data. - This military standard supersedes and includes the requirements of MIL-S-60902(MU), dated 17 August 1967.

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